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UNIT- II NUMBER THEORY

Abstract Algebra and Number Theory

- * Modern Cryptography
- * Plaintext and Ciphertext (generated by encryp. algo)
- * Most encryption is based heavily on number theory
- * Abstract Algebra

Concepts :-

- * The Division Algorithm ✓
- * The Euclidean Algorithm ✓
- * The Extended Euclidean Algo. ✓
- * Modular Arithmetic ✓
- * Groups, Rings, Fields and Finite Fields
- * Polynomial Arithmetic
- * Prime nos ✓
- * Fermat's and Euler's Theorem ✓
- * Testing for primality ✓
- * The Chinese Remainder ✓
- * Discrete Logarithm ✓
- * Finite fields ✓

Divisibility and Division Algorithm :-

⇒ Divisibility

For any two integers, we say "b divides a" or "b/a" if there exist an integer k such that

$$a = kb$$

This means when "a" is divided by "b", then there is no remainder.

Ex:- 4 divides 12 $\rightarrow 4|12$

$$12 = 3 \cdot 4 \quad (a = kb)$$

3 is an integer

$\therefore 12$ is divisible by 4

$$b|a = \frac{a}{b} \quad \begin{array}{l} \text{dividend} \\ \text{divisor} \end{array} \quad \frac{a}{b} = k$$

#

⇒ Properties :-

- If $a|1$, then $a = \pm 1$
- If $a|b$ and $b|a$ then $a = \pm b$ or $a = \pm 1$
- If both $a|b$ and $b|a$, then a & b must be same except their sign.
- Any $b \neq 0$ divides 0
 → Any nonzero no. b divides 0
 → 0 can be written as $b \cdot 0$
 $0 = kb \rightarrow 0 = 0 \cdot b$
 $\hookrightarrow$ integer
- If $a|b$ and $b|c$, then $a|c$
 → Transitivity of division
 $2|4$ & $4|12 \Rightarrow 2|12$
- If $b|a$ and $b|b$, then $b|(ma + nb)$ for arbitrary integers m and n .
 → If b divides both a & b , then b divides the linear combination $(ma + nb)$ for any arbitrary integers m & n .

Linear combination - An expression constructed from a set of terms by multiplying each term by a constant and adding the result, linear combination of x & y is $ax + by$ where a & b are arbitrary constants.

Division Algorithm

It provides a way of expressing the division of any two +ve integers a & b , where $b \neq 0$, in terms of 'a' a quotient and a remainder.

Let 'a' and 'b' be two +ve integers such that $b \neq 0$, then on dividing 'a' by 'b' there exist unique integers q (quotient) and r (remainder) such that

$$a = bq + r$$

where

$$0 \leq r < |b|; q = a/b$$

a - Dividend

b - Divisor

q - quotient

r - remainder

Ex $a=17, b=5$

On dividing 'a' by 'b' we get quotient $q=3$ ($5 \times 3 = 15$) which is the largest multiple of 5 less than or equal to 17.

The remainder $r=2$

By Division Algorithm

$$a = bq + r$$

$$17 = 5 \times 3 + 2$$

$$0 \leq r < 5$$

* Applications

* Modular Arithmetic - The remainder 'r' is the result of $a \bmod b$ which is the foundation of modular arithmetic.

* Greatest Common Divisor - The Euclidean algorithm, used to find the GCD of two numbers, is based on the repeated application of the division algorithm.

* Cryptography - Modular arithmetic and the division algorithm are essential for many cryptographic systems, like RSA, which rely on modular exponential and inverse theory.

Modular Arithmetic

- System of arithmetic for integers
- Wraps around after a certain value called modulus.
- Central mathematical concept in cryptography

→ Congruence

- In cryptography, congruence ($\equiv$) instead of equality ($=$)

ex $15 \equiv 3 \pmod{12}$

12	15
	12
	3

$23 \equiv 11 \pmod{12}$

12	23
	12
	11

$10 \equiv -2 \pmod{12}$

12	10
	0
	10
	(-2)

$$\therefore a \equiv b \pmod{m}$$

i.e.

$$a = km + b$$

$$\begin{array}{r|l} k & a \\ m & b \end{array}$$

→ Modular arithmetic is a kind of integers arithmetic that reduces all no. to one of a fixed set $\{0, 1, \dots, n-1\}$ for some no. Any integer outside this range is reduced to one in this range by taking the remainder after division by n .

⇒ Modular Arithmetic

- If 'a' is an integer and 'n' is a +ve integer, we define a mod n to be the remainder when a is divided by n.

$$11 \pmod{7} = 4$$

$$-11 \pmod{7} = 3$$

$$(-11 \div 7) \times 7 = -15$$

$$-11 - (-15) = 4$$

$$\begin{array}{r} 213 \\ 79 \\ \hline 134 \end{array}$$

$$\begin{array}{r} 213 \\ 78 \\ \hline 104 \end{array}$$

$$\begin{array}{r} 213 \\ 79 \\ \hline 117 \end{array}$$

$$13 \overline{) 13^2}$$

$$\begin{array}{r} 13 \\ \underline{13} \\ 0 \end{array}$$



Two integers 'a' and 'b' are said to be congruent modulo n, if $(a \bmod n) = (b \bmod n)$. This is written as $a \equiv b \pmod{n}$.

$$73 \equiv 4 \pmod{23}$$

$$21 \equiv -9 \pmod{10}$$

Properties:- (a) Congruence

- $a \equiv b \pmod{n}$ if $n \mid (a-b)$
- $a \equiv b \pmod{n}$ implies $b \equiv a \pmod{n}$
- $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$ imply $a \equiv c \pmod{n}$
- $[(a \bmod n) + (b \bmod n)] \bmod n = (a+b) \bmod n$
- $[(a \bmod n) - (b \bmod n)] \bmod n = (a-b) \bmod n$
- $[(a \bmod n) * (b \bmod n)] \bmod n = (a*b) \bmod n$

Q find $11^7 \bmod 13$

$$11^2 = 121 \equiv 4 \pmod{13}$$

$$11^4 = (11^2)^2 \equiv 4^2 \equiv 3 \pmod{13}$$

$$11^7 = 11 \times 11^2 \times 11^4$$

$$11^7 \equiv 11 * 4 * 3$$

$$\equiv 2 \pmod{13}$$

$$\begin{array}{r} 9 \\ 13 \overline{) 121} \\ \underline{117} \\ 4 \end{array}$$

$$\begin{array}{r} 1 \\ 13 \overline{) 16} \\ \underline{13} \\ 3 \end{array}$$

$$\begin{array}{r} 11 \\ 11 \\ \hline 132 \end{array}$$

Properties of Modular Arithmetic

① Commutative laws

$$(a+b) \bmod n = (b+a) \bmod n$$

$$(a \times b) \bmod n = (b \times a) \bmod n$$

② Associative laws

$$[(a+b)+c] \bmod n = [a+(b+c)] \bmod n$$

$$[(a \times b) \times c] \bmod n = [a \times (b \times c)] \bmod n$$

③ Distributive laws

$$[a \times (b+c)] \bmod n = [(a \times b) + (a \times c)] \bmod n$$

④ Identities

$$(0+a) \bmod n = a \bmod n$$

$$(1 \times a) \bmod n = a \bmod n$$

⑤ Additive Inverse

for each $a \in \mathbb{Z}_n$, there exists a $-a$ such that

$$a + (-a) \equiv 0 \bmod n$$

Prime Numbers

- A prime no. is a no. greater than 1 with only two factors i.e. 1 and no. itself

Composite no. - which are not prime no. i.e. having factors other than 1 and no. itself

→ Facts & Properties :-

- Divisibility - A prime no. N can only be divided completely (no/o remainder) by 1 and N itself.
- 2 is the smallest prime no. and also the only even prime no. (because other even no. is divisible by 2)
- 1 is not a prime no.
- Except for 2 and 5, all prime no. ends with 1, 3, 7 and 9
→ only 2 & 5 are prime no. - all no. ending with 2 or 5 are composite.

* Applicatn

i) Building Blocks of Number

Every natural no. greater than 1 can be expressed as a product of prime no.

→ This is called fundamental theorem of Arithmetic.

No.

6

12

100

Prime

$2^1 \times 3^1$

$2^2 \times 3^1$

$2^2 \times 5^2$

Factoriztn

Prime no 2, 3

2, 3

2, 5

ii) Cryptography

- Many encryptn alga. are based on prime nos like RSA, which is used to secure interest communicatns

- It is very fast to multiply two large prime nos

Q.1 Explain different methods of modular arithmetic?
Q.2 What is modular arithmetic and give a simple example?

⇒ Methods of Modular Arithmetic :-

① Repeated Multiplication :

→ For small exponents, the simplest method is to repeatedly multiply the base modulo m .

② Exponentiation by Squaring :

→ A more efficient method, especially for large exponents. It involves repeatedly squaring the base and multiplying the results based on the binary representation of the exponent.

③ Chinese Remainder Theorem (CRT) :

→ Used to solve systems of linear congruences. It states that if $m_1, m_2, \dots, m_n$ are pairwise coprime integers, then for any integers $a_1, a_2, \dots, a_n$, there exists a unique solution x modulo $M = m_1 * m_2 * \dots * m_n$.

⇒ Key concepts in modulus arithmetic :-

① Modulus :- The modulus is a value of which no's wrap around

Ex:

$$1 \bmod 12 = 1$$

$$12 \bmod 12 = 0$$

$$2 \bmod 12 = 2$$

$$13 \bmod 12 = 1 \longrightarrow \text{calculation } 13 \times 13 =$$

$$13 - 13 \div 12 \times 12$$

In calc

$$\boxed{\begin{array}{l} \text{for } A \bmod B \\ A - A \div B \times B \end{array}}$$

② Congruence - Two no's are said to be congruent if they have same remainder when divided by modulus

③ $38 \equiv 14 \pmod{12}$

$$\boxed{38 \bmod 12 \equiv 14 \bmod 12}$$

$$38 \bmod 12 \Rightarrow 38 = 12 \times 3 + r$$

$$38 = 36 + r$$

$$\boxed{r = 2}$$

Q $35 \equiv 15 \pmod{10}$

$$35 \pmod{10} \Rightarrow 35 = 10 \times 2 + r$$

$$35 = 20 + r$$

$$\boxed{r = 15}$$

Q $5x \equiv 4 \pmod{3}$

$$4 \pmod{3} \equiv 1 \pmod{3}$$

$$5x \equiv 1 \pmod{3}$$

$$10 \times 2 \equiv 1 \pmod{3}$$

finding multiplicative inverse of mod 3

$$1 \pmod{3} = 4, 7, 10 \dots$$

$$5x \equiv 1 \pmod{3}$$

$$5x \equiv 4$$

$$x = 4/5$$

$$5x \equiv 10$$

$$x = 10/5 = 2 \dots 1 + 5 + 10 + 15 \dots$$

Euclidean algorithm :-

- It is the most effective method for computing the GCD of two integers.
- It is based on the principle that the GCD of two no.s does not change if the larger no. is replaced by its remainder when divided by smaller no.
- To find $\text{GCD}(a, b)$ such that $a > b$
We can repeatedly replace a with b and b with the remainder of a/b until the remainder is 0.
The last non-zero remainder is the GCD of a & b .

Ex 1 $\text{GCD}(12, 33)$

$$33 \div 12 = 2$$

$$33 = 12 \times 2 + 9 \quad \therefore 33 \div 12 = 2$$

$$12 = 9 \times 1 + 3$$

$$9 = 3 \times 3 + 0$$

$$\therefore \text{GCD}(33, 12) = 3$$

Ex 2 $\text{GCD}(48, 18)$

$$48 > 18$$

$$48 = 18 \times 2 + 12$$

$$18 = 12 \times 1 + 6$$

$$12 = 6 \times 2 + 0$$

$$\therefore \text{GCD}(48, 18) = 6$$

Ex 3 Fermat's little theorem :-

If 'p' is a prime no. and 'a' is a +ve integer not divisible by 'p' then

$$a^{p-1} \equiv 1 \pmod{p}$$

Ex $p=5, a=2$

$$a^{p-1} \equiv 1 \pmod{p}$$

$$2^{5-1} \equiv 1 \pmod{5}$$

$$2^4 \equiv 1 \pmod{5}$$

$$16 \equiv 1 \pmod{5}$$

$$\begin{array}{r} 3 \\ 5 \overline{) 16} \\ \underline{15} \end{array}$$

Therefore, Fermat theorem holds true!
for $p=5$ & $a=2$

Q.2

$$p=13, a=11$$

$$a^{p-1} \equiv 1 \pmod{p}$$

$$11^{12} \equiv 1 \pmod{13}$$

$$11^2 \equiv 1 \pmod{13}$$

(11)

$$(11)^2 \times (11)^2 \times (11)^2$$

$$4 \times 4 \times 4 \equiv 1 \pmod{13}$$

$$64 \equiv 1 \pmod{13}$$

$$- 2^{12} \equiv 1 \pmod{13}$$

$$- 2^{4 \times 3} \equiv 1 \pmod{13}$$

$$3^3 \equiv 1 \pmod{13}$$

$$2^7 \equiv 1 \pmod{13}$$

Q.1
Euler's Theorem :-

For every +ve integers 'a' & 'n' which are said to be relatively prime, then

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

Q.1 $a=3, n=10$

$$3^{\phi(10)} \equiv 1 \pmod{10}$$

$$\phi(10) = 4$$

$$\begin{aligned} \phi(10) &= 2 \times 5 - 1 \\ &= (2-1)(5-1) \\ &= 1 \times 4 \end{aligned}$$

$$3^4 \equiv 1 \pmod{10}$$

$$81 \equiv 1 \pmod{10}$$

Q.2 $a=2, n=10$

$$2^{\phi(10)} \equiv 1 \pmod{10}$$

$$2^4 \equiv 1 \pmod{10}$$

$$16 \equiv 6 \pmod{10}$$

It does hold for Euler's Theorem

Groups and Abelian gp

- A group G denoted by $\{G, \cdot\}$, is a set under some operation $(\cdot)$ if it satisfies the CAN properties

C - closure

A - Associative

I - Identity

N - Inverse

- A gp is said to be Abelian if it already a gp and commutative property is also satisfied i.e.
 $(a \cdot b) = (b \cdot a)$ for all a, b in G

Property

Explanation

Closure

$a, b \in G$, then $(a \cdot b) \in G$

Associative

$a \cdot (b \cdot c) = (a \cdot b) \cdot c$ for all $a, b, c \in G$

Identity element

$(a \cdot e) = (e \cdot a) = a$ for all $a, e \in G$

Inverse element

$(a \cdot a') = (a' \cdot a) = e$ for all $a, a' \in G$

Commutative

$(a \cdot b) = (b \cdot a)$ for all $a, b \in G$

Ex 2s (2.7) y/p

Let $G = \{1, 2, 3, 4, 5, 6, 7\}$

① Closure

$1, 2 \in G$, then

$$(1+2) = 3 \in G$$

$\therefore$ closed under $+$

② Associative

$1, 2, 5 \in G$

$$11(2+5) = (1+2)+5 = 11 \in G$$

$\therefore$ hence, true

③ Identity element

$$(5+0) = (0+5) = 5 \text{ for all } a \in G$$

④ Inverse element

$$(5 + (-5)) = ((-5) + 5) = 0 \text{ for all } a, a' \in G$$

⑤ Commutative

$$5+4 = 4+5 = 9 \in G$$

$N \rightarrow$ Set of all natural no.s
 $W \rightarrow$ Set of all whole no.s
 $Z \rightarrow$ Set of all integers
 $C \rightarrow$ " " complex no.s
 $Q \rightarrow$ " " rational no.s
 $R \rightarrow$ " " real no.s
 $Z^+ \rightarrow$ " " +ve integers
 $Z^- \rightarrow$ " " -ve integers

Q Describe computations in finite fields and their relevance to cryptography?

Ans. Finite fields are mathematical structures that are used extensively in cryptography due to their properties and computational efficiency.

A finite field is a set of elements with defined addition and multiplication operators that satisfy certain properties.

$\Rightarrow$ Key properties of finite fields :-

① Closure

② Associativity

③ Commutativity

④ Identity elements

⑤ Inverse elements

⑥ Distributivity

→ Computations in finite fields:-

① Addition - The addition operation in a finite field is typically performed modulo the field's characteristic.

② Multiplication - "The multiplication operation"

③ Exponentiation - Exp. in finite fields can be efficiently computed using algorithms like square-and-multiply method.

④ Division - Division in finite field is equivalent to multiplying by multiplicative inverse of the denominator.

→ Relevance in Cryptography:-

① Elliptic Curve Cryptography (ECC)

② Diffie-Hellman Key Exchange

③ RSA Cryptography

Ques

Q Discuss the properties that are satisfied by Groups, Rings, & fields?

Ans. • Groups, rings, and fields are fundamental algebraic structures that are widely used in mathematics, computer science and other fields.
• They are defined by specific sets of properties that govern the operations performed on their elements.

⇒ Groups - a gp is a set G equipped with a binary operation $*$ that satisfies the following properties:

- Closure - for all $a, b \in G$, $a * b \in G$
- Associativity - for all $a, b, c \in G$, $(a * b) * c = a * (b * c)$
- Identity element - there exists an element $e \in G$ such that for all $a \in G$, $a * e = e * a = a$
- Inverse element - for every $a \in G$, there exists an element $a^{-1} \in G$ such that $a * a^{-1} = a^{-1} * a = e$

⇒ Rings - a ring is a set R equipped with two binary operators, usually denoted $+$ and $*$, that satisfy the following properties :-

- R is a group under addition
- R is associative under multiplication
- for all $a, b, c \in R$, $a * (b + c) = a * b + a * c$
and

$$(a + b) * c = a * c + b * c$$

[distributive property]

⇒ Fields - a field is a set F equipped with two binary operators, $+$ and $*$, that satisfy the following properties :-

- F is a ring
- F is commutative under multiplication
- for every non-zero element $a \in F$, there exists a multiplicative inverse $a^{-1} \in F$ such that $a * a^{-1} = a^{-1} * a = 1$

Primality Test

Fermat's little theorem: If 'n' is a prime number, then for every a, $1 \leq a \leq n-1$

$$a^{n-1} \equiv 1 \pmod{n}$$

Randomized Property - if a given no. is prime, then this method always returns true.

If the given no. is composite (or non-prime), then it may return true or false, but the probability of producing incorrect results for composite is low and can be reduced by doing more iterations.

→ Algorithm

1) Repeat following 'k' times:

- Pick 'a' randomly in the range $[2, n-2]$
- If $\gcd(a, n) \neq 1$, then return false
- If a^{n-1} not equivalent to $1 \pmod{n}$, then return false

2) Return true [probably prime]

Ex:

Ex: for $n=7$, we will check $a^6 \pmod{7}$ for several values of a .

① $a=2$

$2^6 \pmod{7} = 64 \pmod{7} = 1$, so 7 might be prime

② $a=3$

$3^6 \pmod{7} = 729 \pmod{7} = 1$, so - "

③ $a=5$

$5^6 \pmod{7} = 15625 \pmod{7} = 1$, so -

Ex 2 for $n=15$, check $a^{14} \pmod{15}$

① $a=2$

$2^{14} \pmod{15} = 16384 \pmod{15} = 4$, so 15 is comp

② $a=3$

$3^{14} \pmod{15} = 4782969 \pmod{15} = 9$, so, 15 - "

Ex 3 Carmichael Number (composite no. with probabilistic nature)

→ A Carmichael no. is a composite no. that passes Fermat's test for most values of 'a' making it appear prime under the test.
let's use $n=561$, which is a Carmichael no.

for $n=561$, check $a^{560} \pmod{561}$

- ① for $a=2$
 $2^{560} \pmod{561} = 1$, so 561 might be prime
- ② for $a=5$
 $5^{560} \pmod{561} = 1$, so 561 is likely prime

Complexity

→ TC: $O(k \log n)$, Note - power function takes $O(\log n)$ time

→ SC: $O(\min(\log a, \log b))$

Miller Rabin Method

- The Miller-Rabin Primality Test is an improved version of Fermat's test that provides a more robust probabilistic method for checking whether a no. is prime.
- While it still cannot guarantee primality in every case (unless all bases are checked, which can be computationally expensive), it has a much lower chance of false positives (i.e., mistakenly identifying a composite no. as prime).

→ Algorithm

- ① Write $n-1 = 2^s \cdot d$:
compute 's' and 'd' such that $n-1 = 2^s \cdot d$,
where 'd' is odd
- ② Random Base a:
Pick a random integer a b/w 1 and $n-1$
- ③ Modular Exponentiation
compute $x = a^d \bmod n$
- ④ Checking conditions:
→ If $x = 1$ or $x = n-1$, the test is inconclusive
and 'n' is probably prime.

→ otherwise, square x , and repeat the process $s-1$ times

- If any $x = n-1$, the test is inconclusive, and ' n ' is probably prime.
- If none of the sq. give $x = n-1$, then ' n ' is composite.

ex. $n=17$

①

$$17-1 = 2^4 \cdot 1$$

$$s=4, d=1$$

② $a=3$

③ $x = 3^1 \bmod 17$

④ check x

Since $x \neq 1$ and $x \neq 16$, we proceed by squaring x

→ square x

$$x = 3^2 \bmod 17 = 9 \bmod 17 = 9$$

$$x = 9^2 \bmod 17 = 81 \bmod 17 = 13$$

$$x = 13^2 \bmod 17 = 16$$

• Since $x=16$, the test is inconclusive and 17 is probably prime.

$$n=39$$

$$39-1 = 2 \cdot 19$$